

Privacy Threats of Behaviour Identity Detection in VR

Dilshani Kumarapeli*
HIT Lab NZ
University of Canterbury

Sungchul Jung†
Game Design and Development
Kennesaw State University

Robert W. Lindeman‡
HIT Lab NZ
University of Canterbury

ABSTRACT

Modern VR technology generates large volumes of data by attaching various sensors around the user. Over time, researchers have used these data to create unique behavioural keys to provide authentication to immersive applications. However, these approaches come with various behavioural privacy risks. Hence, through this work, we investigate the privacy risks associated with behavioural identity detection in VR and how users' physical appearance affects this behavioural detection. We found that users could be identified across various sessions with 80% accuracy once tracked, and that physical appearance did not impact the behaviour detection.

Keywords: Behavioural Privacy, Identity Detection, Physical Factors

1 INTRODUCTION

One of the more-recent developments in VR authentication is to use a person's behaviours as an authentication medium [6]. Due to the nature of VR, traditional approaches like entering a password or PIN with a keyboard need to be replaced by new, less efficient input methods, such as air tapping keyboards, rendering conventional authentication procedures inefficient and unattractive [1]. Therefore, behaviour-based identity detection systems are an affordable and efficient choice to provide identification and authentication in VR since these algorithms can identify people using their behavioural actions, such as eye-gaze patterns or head movements, without requiring them to enter any textual information. Likewise, these algorithms could be used to authenticate users repeatedly during the session without interfering with the immersive experience [8]. However, like every other approach, despite their advantages, these approaches also have certain drawbacks regarding user privacy [5]. VR's capability to obtain eccentric user behaviour data is the primary cause of these privacy concerns [9, 11]. Traditional passwords and PINs are knowledge-based authentication systems. Hence, if these passwords or PINs are somehow discovered or revealed, identity can be regained efficiently by changing or reissuing them [2]. However, since behavioural identity is built upon the user's natural behaviour, it is hard to change them once they are revealed, and therefore have permanent consequences. Behavioural identity is more strongly coupled to the actual user than traditional authentication methods [13].

Through this research, we aim to draw attention to the behavioural privacy issues that the aforementioned authentication systems may cause.

- RQ_1 : Can user behaviour data be used to identify particular users performing different tasks in VR?
 - H_{11} : Machine learning based identity detection algorithms will be accurate when the training data and testing data are from the same task.

*e-mail: kumarapeli.kumarapeli@pg.canterbury.ac.nz

†e-mail: sjung11@kennesaw.edu

‡e-mail: gogo@hitlabnz.org

- H_{12} : Machine learning based identity detection algorithms will be accurate when the training data and testing data are from different tasks.

- RQ_2 : Does the user's appearance (e.g., skin tone, presence of facial hair) impact identity detection accuracy?

- H_{21} : There is no significant influence of user skin tone on machine learning based identity detection accuracy.
- H_{22} : There is no significant influence of user hand dominance on machine learning based identity detection accuracy.
- H_{23} : There is no significant influence of user facial hair presence on machine learning based identity detection accuracy.

2 METHODOLOGY AND STUDY DESIGN

Our user study was approved by the Human Research Ethics Committee of our university. We investigated our research questions using a 2 x 1 within-subjects experimental design, with one independent variable "Identity Accuracy" with two levels "Reading" and "Speaking." The study consisted of three different stages: **R1**: Read the provided bedtime story aloud, **S1**: Retell the R1 story in the participants' own words, and **R2**: Read a separate bedtime story. During the experiment, we automatically logged 73 behavioural data points in terms of eye gaze, eye movement, lip movement, hand and head movement data, and collected self-reported responses, including skin tone (using Fitzpatrick Scale [4]), dominant hand, and presence of any facial hair, at the end of the study session using a Qualtrics questionnaire. Figure 1 shows the application interface we created to collect the data.

We then trained the classification algorithms described in Table 1 using the **R1** dataset and tested using the **R2** and **S1** datasets to analyse RQ_1 . Next, RQ_2 was analysed using the classification results of the model with the highest overall accuracy.

3 RESULTS

A total of 40 participants (Gender: Male: 18, Female: 21, Demi-Boy: 1; Age: $M = 28.2$, $SD = 7.46$) were involved in this study.

3.1 RQ_1 : Identify Particular Users in VR

We completed training the identity detection model using the classification algorithms described in Section 2 with the **R1** dataset. With the trained model, we tested the **R2** and **S1** datasets. The **R2** dataset consists of data gathered by performing the same task ('Reading') as the **R1** dataset to analyse H_{11} , and the **S1** dataset ('Speaking') consists of data gathered by performing a slightly different task than the **R1** dataset to analyse H_{12} . We provide each classification model's training and testing accuracies respectively, in Table 2. The results indicate that Medium KNN and Single Layer Neural Network show the highest accuracy in **R2** and **S1**.

3.2 RQ_2 : Personal Appearance Attribute Impact

We tested hypotheses H_{21} (Skin Tone), H_{22} (Hand Dominance) and H_{23} (Facial Hair) with the **R2** dataset. First, we conducted an independent-samples Kruskal-Wallis test, which showed that there

Classification Model	Parameters Used
Fine Tree [10]	Maximum number of splits: 100 Split criterion: Gini's diversity index Surrogate decisions splits: Off
Medium KNN [3]	Number of neighbours: 10 Distance metric: Euclidean Distance weight: Equal Standardize data: True
Bagged Trees [7]	Ensemble method: Bag Learner type: Decision Tree Maximum number of splits: 1,208,207 Number of layers: 30
Single Layer Neural Network [12]	Number of fully connected layers: 1 First layer size: 25 Activation: ReLu Iteration limit: 1,000 Standardize data: True
Bilayered Neural Network [12]	Number of fully connected layers: 2 First layer size: 10 Second layer size: 10 Activation: ReLu Iteration limit: 1,000 Standardize data: True

Table 1: Parameters used for the classification algorithms.

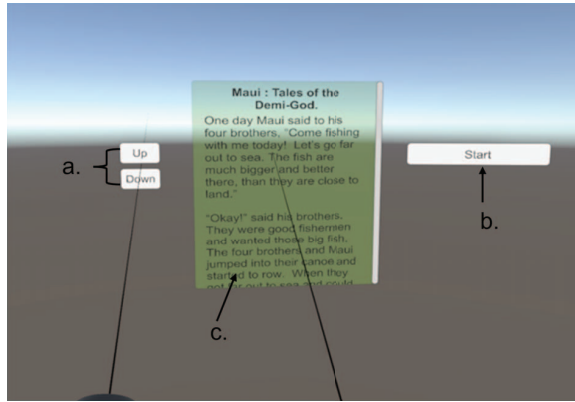


Figure 1: The Story Reading Interface. (a) The “Up” and “Down” buttons helped to scroll through the story. (b) The “Start” button toggled into a “Stop” button once started. (c) The story reading interface could be controlled using “Up” and “Down” buttons.

was no significance impact over the behavioural identity detection accuracy from any of these variables (Skin Tone: $h(4) = 6.6657$, $p = 0.115$; Hand Dominance: $h(1) = 0.926$, $p = 0.336$; Facial Hair: $h(1) = 2.397$, $p = 0.122$). Hence, we accepted the null hypotheses for these variables.

4 CONCLUSION

This paper examined the underlying privacy dangers that behavioural identity detection technologies in VR potentially pose. First, we demonstrated that even simple classification approaches could identify the participants with a high detection rate utilising behavioural features in VR. Following that, we investigated how reliably these trained classifiers can identify the same person performing different tasks. Finally, we investigated the impact of the user's physical attributes on the classification of behavioural identity. These findings highlight the significance of offering greater privacy protection tools for VR users to benefit both VR consumers and the VR industry considering the fact that privacy should be seen as a right, rather than a privilege. To the best of our knowledge, this is the first study to investigate the link between physical characteristics and the accuracy of behaviour identity recognition through a formal user study. We

Model Name	R1 Accuracy	R2 Accuracy	S1 Accuracy
Fine Tree	92.7	41.7	60.1
Medium KNN	99.9	86.6	76.8
Bagged Trees	100.0	55.0	71.1
Single Layer Neural Network	100.0	82.7	79.4
Bilayered Neural Network	100.0	74.9	68.2

Table 2: Training and testing accuracies for the classification models.

hope the results we present here spawn new interest and significant effort into analysing behavioural techniques to protect VR users better.

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